

Persona Co-Design for Improving Digital Accessibility

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ABSTRACT

This paper examines approaches to developing authentic accessibility personas that can help improve the design of products for persons with disabilities. Personas have been a controversial empathy-building tool that have been shown to have potentially harmful effects on the design of accessible products when they are not well-made or used appropriately. In this study we describe a nested co-design approach that involves personas as product and tool, and that includes persons with disabilities at multiple stages and in multiple functions within the design process. These approaches are used in the context of the improvement of digital accessibility teaching and learning materials at higher education institutions.

CCS CONCEPTS

• **Human-centered computing** → *Systems and tools for interaction design*;

KEYWORDS

Personas, Co-Design, Digital Accessibility, Higher Education

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1 INTRODUCTION

Most people know too little about accessibility to create products and services that are accessible to all, and conform to the relevant

regulatory guidelines [23–25]. While there has been a marked increase in the availability of information on digital accessibility, this remains limited. Not everyone is able to take part in training sessions or have regular access to accessibility experts. One way of mitigating this issue is to teach people about accessibility using personas. Personas are fictional characters that help the designer to empathise with their user and see the world through their eyes [3, 4]. In the context of accessibility, designers are made aware of barriers people with access needs face, as well as the assistive technologies and strategies they use to overcome these barriers. Personas are a double-edged sword in that they are created with the goal of creating empathy and understanding for the user by increasing perspective-taking but they often fall short of this goal. Poorly-designed personas can reinforce stereotypes and tropes rather than increase empathy. This is often the case with lean personas, and personas that are not based on thorough user research and co-design principles [9, 13].

This issue is particularly problematic when designing accessibility personas. There is a clear danger of reinforcing stereotypes, and showing persons with disabilities (PwD) as less capable than they actually are. The challenge is therefore: *How do we design good accessibility personas?* This paper examines proposed approaches to answer this question. The context of the experiment is the higher education setting, where lecturers are designers of teaching and learning materials, and students with disabilities are the end-users. This late-breaking work examines these issues in the context of an ongoing project for improving digital accessibility in higher education in Germany. The data for this study were collected as part of Hochschul-Initiative Digitale Barrierefreiheit für Alle (SHUFFLE) project which has an overall aim of improving digital accessibility at the German tertiary level. One approach to achieving this goal is by improving the technical and didactic competencies of teaching staff, and by improving the accessibility of the technological infrastructure for teaching and learning.

The "BlindDate" sub-project involves the creation of a digital platform through which teaching staff can encounter interactive personas of students with individual needs (SmiBs). The central focus was the design of authentic accessibility personas of students

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with a range of barriers and access needs. One approach is the use of a **co-design methodology** that involves PwD at all the product design stages, including them as experts as well as designers, with core roles in iterative feedback and evaluation. A second approach is conceptualizing and using personas as a product in a **nested design space**. In this approach, the persona is not a single chunk of information, but a collation of persona artefacts, including additional media, “a day in the life” stories, quotes, simulations, and links to additional material. In the following sections, relevant background literature and methodological choices in persona design are discussed, and a description of the current progress on the development of eight planned persona artefacts is provided, concluding with insights into future work.

2 RELATED LITERATURE

2.1 Personas in Design

Well written personas should be able to help designers better empathize with users, their challenges and strategies in different scenarios in which the product or service may be used. This makes them potentially very helpful in user-centric decision-making [17, 26]. While there has been a degree of concern about the efficacy of personas in the design process, these have been mainly ascribed to how the personas themselves are designed, how they are evaluated for credibility and goodness of fit, and how they are actually used in the design process [9, 20]. Persona development takes place in different ways, depending on the time and resources available to the design team and the goals for persona use. Quick development in agile teams often involves a largely qualitative approach based on data derived from user interviews and other ethnographic methods. With the increased availability of data from user analytics, there has been an increase in quantitative approaches to developing personas. While qualitative approaches use relatively little data [19], quantitative approaches can use large-scale datasets to cluster user types [1] which can be used as a basis for persona development.

2.2 Persona Representations of Disability

Personas of people with access needs and user scenarios involving these personas, have been used as a method for integrating accessibility considerations into the design process [21]. These user personas and related scenarios in design use, have received limited attention in the literature, but can be found across domains including home appliances [12], healthcare [22] and mobile applications [18]. However, developing authentic, non-stereotypical personas of PwD has been challenging [9, 13], especially when multiple intersections of disability and identity are considered [6]. The central aim in persona development in these cases has been to present PwD in a manner that is both ethically responsible, as well as authentically representative of their experiences.

This goes in hand with cognitive-affective goals to develop empathy within designers, in order to better inform their design decisions [8]. This empathy-building aspect is even more critical when users have disabilities which may directly and indirectly affect their usage of the product. There has been some criticism of the effectiveness of disability personas for empathy-building in the design context [2, 10, 14]. Failures of authenticity appear to arise from two key related areas within the design process: (i) displacing the voices

of those with disabilities; and (ii) separation of roles, where the designers are non-disabled, and the PwD have limited or no roles as designers. This exclusion at multiple stages in the design process is not conducive to developing products that truly address the needs of PwD.

2.3 Persona Co-Design

Co-design is an approach that draws on collective stakeholder creativity in the design process [16]. The aim is to include persons with the most knowledge and/or experience with the required needs in the design process. This approach is even more crucial when products and services for PwD are involved, due to the problems outlined in the previous section. Early research in this area [21] focused on standard human-centred design principles in data collection and synthesis. One aspect of this approach is the use of persona workshops which include PwD in the creation of personas with disabilities. This research often stressed having someone in design teams with knowledge/or experience of disabilities. One study [15] used similar methods in the creation of co-designed personas for persons with Parkinson’s disease, dementia or aphasia and highlighted their use in developing assistive software and hardware. Another study [7] extends these methodologies by including PwD in various stages of the development cycle including workshops, prototyping, and usability testing. The authors argued that this would help offset some of the negative aspects of disability personas, while taking advantage of the strengths inherent in the approach.

3 METHODS

In this section, we describe some of the methodological approaches and decisions involved in the development of personas for improving digital accessibility in the higher education domain.

3.1 Surveys

Initial data collection focused on two surveys that were administered to 179 teaching staff and to 695 students with disabilities and/or other types of barriers such as family care needs or chronic illness. Respondents were recruited through opportunity sampling, in which an email was sent to all the students and staff members at each of the project universities. The survey was piloted on a small-scale at one of the institutions, and evaluated for accessibility. Each survey was administered online using Lime Survey, and included 17 elements that included Yes-No questions, Likert items, and open-ended responses. The teaching staff came from six German universities in three states, and included professorial and non-professorial faculty. The survey sought the following information: knowledge and understanding of digital accessibility, their contact with students with various access needs, their current methods in producing learning materials, using the Learning Management System (LMS), running exams, and their attempts (or not) of making their teaching more accessible.

The students surveyed came from eight universities in four states, with 67% of them registered for Bachelor degrees, and with their then current length of enrollment ranging from the first to sixteenth semesters. The information sought from the students focused on: types of disability and/or type of barriers experienced, specific

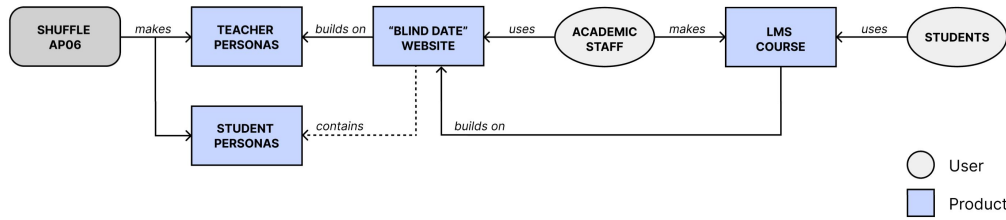


Figure 1: Workflow in the Project Design Space

information on their experience with learning materials, online teaching, exams, and LMS navigation. Key statistics, themes and quotes identified in the thematic analysis were copied to the collaborative online platform, Miro, for affinity mapping and further synthesis activities.

3.2 Interviews

Semi-structured interviews have been conducted with both SmiBs and teaching staff on an ongoing basis from March 2022. All interviews were audio-recorded with the consent of the participants and later analyzed thematically by the design team. By February 2023 fifteen in-depth student interviews have been completed, with the aim of gaining a better understanding of their past and current challenges in their university courses. This included what strategies they used to try to obviate the effects of the limited or non-existent digital accessibility solutions. The students in these interviews were chosen due to having access needs that were not well represented in the survey data, or where more detailed information on a topic was required. Students were asked about their understanding of digital accessibility, as well as specific instances of problems with the learning environment, learning materials and examinations. They were also asked about their attempts to find solutions themselves or with the assistance of fellow students, teaching or administrative staff.

During the same period, eight teaching staff interviews were completed with faculty that did not take part in the survey. Since teaching faculty generally provided quite detailed responses to the open-ended survey questions, these interviews were held to clarify the consistency of key issues and stances raised, and to determine if there were any issues not highlighted in the survey. These interviews centred on the lecturers' knowledge and understanding of digital accessibility, and their awareness and sensitivity to the needs created by different barrier types. In addition, the lecturers provided examples of their own attempts to have more inclusive digital learning environments, or problems they encountered that prevented them from doing so. Key themes, strategies and quotations from both sets of interviews were added to the Miro board for further analysis.

3.3 The Design Space

The project design space is based on co-design, which is a participatory design methodology that involves collaboration in the design process between all product stakeholders. This is done in order to

best capture and mobilize their varied experiential and conceptual knowledge. When fully implemented, co-design can be viewed as a user empowerment approach [11], in which users are involved at all stages of the development process: designing, iterative feedback, testing and analysis. In this project the co-design method is utilised in the creation of the disability personas and related content. This methodology is realised within a nested design space [5], in which there is not a single level of users, but direct and indirect users, and direct and indirect products (Figure 1). In this project the teaching staff are the direct users who will be exposed to the interactive personas on the website, and use the given resources to design their classes for improved accessibility.

In this model, the teaching staff are designer-users, who will use the accessibility personas alongside didactic methods like Universal Design for Learning (UDL) to produce more accessible course content for their students. These personas are created by the core design team using the typical persona development methodologies: surveys, interviews and design workshops. In this approach, these personas are both design tools and a product to be used by an intermediate user: the lecturer.

To initialize the design phase, the project team organised two workshops which used Design Thinking methods. The first workshop took place in December 2021 and included project members in their roles as teaching staff, students, developers, designers and researchers. Some of these project members were themselves PwD. Through a series of activities, the team was able to identify pain points, define the problem spaces and their intersections for the two sets of users, and think innovatively about possible solutions that could be implemented through the proposed digital platform. A second design thinking workshop targeted at SmiBs was held in February 2022. The workshop involved nine students with a range of disabilities, and three project members who were themselves PwD and/or experts in educating students with specific disabilities. The number was kept relatively small due to the nature of the activities and the pandemic protocols in place at that time.

Working in small groups, participants described and discussed issues from the student perspective. Themes included situations and implementations that worked well to reduce barriers in their learning, and those that did not work well, or maintained or increased barriers to successful learning. Participants then envisaged what a "perfect world" would look like for them in terms of studying with their disability. From thinking about these questions, and

Table 1: Hannah's Persona Elements

Technology & Strategies	Videos, movies and TV series can be made accessible through automatically generated or manually added subtitles. Sign language translation is equally important for some DHH students.
Lectures	The mouth image of the person giving the lecture should also always be clearly visible, as this helps some people who are DHH to understand. Even under the best conditions (good lighting, not obscured, clear pronunciation) and with trained "lip readers", usually only a maximum of about 30% of speech can be understood.
Learning Materials	Automatic subtitles for videos are now common, but they often have a lot of errors. People who are DHH should have access to more than just what is being said. You should identify the person speaking in a video, and explain any background sounds so that no information is lost.
Communication & Interaction	Group situations in which a lot of people are talking to each other can be very stressful. If the microphones aren't passed on or if people are talking at the same time, information can't be easily extracted in group discussions. This makes participation difficult and people who are DHH can quickly feel excluded.
Examinations	In some cases students can request accommodations for exams, especially oral exams, so that the exam questions are clearly understood.
Quotes	"I would like to see subtitles everywhere without having to ask for them. This should be standard for short videos and longer movies. Also, more patience and understanding when something is not understood acoustically would be really, really great".

exchanging experiences with other participants, the group was then asked to design proto-personas for students with disabilities.

The participants were given four goals in their instructions: (i) to make the persona authentic to their disability experience; (ii) to include strategies and assistive technology used; (iii) to describe barriers and show possible solutions; and (iv) to have the persona make recommendations for lecturers. These personas were further refined following feedback from other participants, either directly or within a role-playing session where they were asked questions from the point of view of the persona's lecturer. Information and artefacts from these design workshops, were then incorporated with data elicited from the survey and the interviews, as the project moved to the next phase.

3.4 Persona Development

From this point, the main design work was carried out by the design team. This consisted of a core team of user experience designers and researchers, some of whom were PwD. An extended project team, which included PwD, also provided feedback. Another level of feedback was provided by project-external students with disabilities and pedagogical experts.

First the core team used survey and interview data to iteratively develop teaching staff personas to represent the target users of the product. These teacher personas served as a design tool for the development of the student personas, as well as for the design of the platform elements for the planned website. While conceptualizing

design choices for student persona artefacts, the possible responses of the teaching personas are considered with questions like "*How would they respond to this persona's needs?*", "*What do they need to know about the disability/the assistive devices/the strategies?*", "*What activities would they take part in or not take part in?*" etc. This allowed the design team to better imagine the design encounter that would take place as teaching staff later interact with the student personas.

An interactive co-design process was applied to the development of the student personas, with quantitative and qualitative data synthesized with the persona prototypes developed in the workshop. This was used to create persona drafts that were modified following different levels of feedback. The process of developing authentic disability presentations in the personas included two main aspects that were regarded as highly important: (i) *multi-stage co-design* and (ii) *targeted persona elements*. The emphasis on *multi-stage co-design* highlights the requirement that co-design with PwD should take place at all levels and stages of the development process, not only at what frequently occurs, the initial data collection and final evaluation stages.

Students with disabilities were, as described in Section 3, involved in interviews, surveys and the design workshop. The core design team, as well as the wider project team included PwD who either currently or previously had themselves been students navigating the university experience. While it was not possible to have the range of all disabilities present for all work sessions, there was

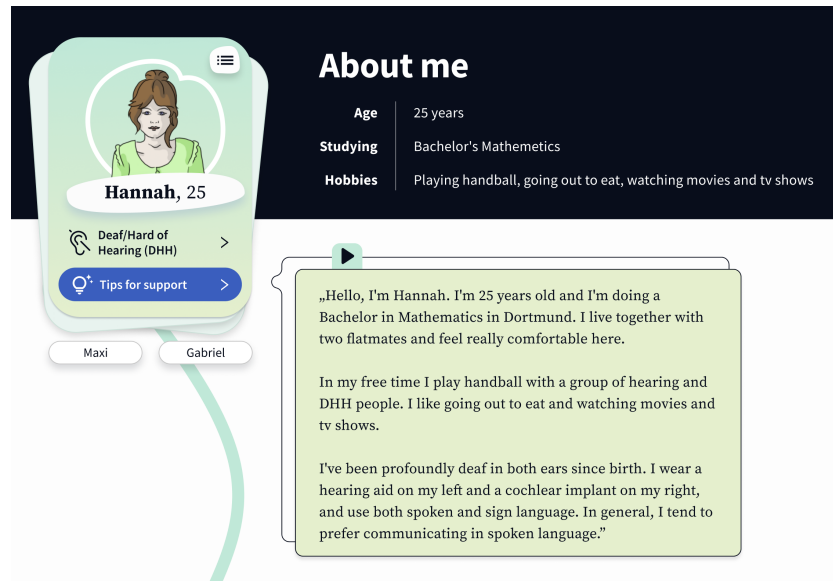


Figure 2: Prototype of the Hannah Persona

an increased facility in having informal access to quick feedback, suggestions and experiential knowledge. More formally, the feedback loop during final persona iterations included two groups that were not part of developing the initial persona drafts: (i) pedagogic experts, and (ii) project-external students with disabilities. The pedagogic experts mainly included teaching staff who specialized in the education of persons with disabilities. A sub-group of pedagogic experts that provided feedback were students who were engaged in graduate-level training in these pedagogic areas, some of whom had disabilities themselves. Additionally, for each persona, the near-final iteration was shown to students with the same or similar disability to provide feedback on the authenticity of the persona's experience, and to make any suggestions for improvement.

Another key aspect in the development of the personas was deciding what *persona elements* to include. The goal with regard to persona elements was not only to show the barriers faced by the students and the challenges faced, but to emphasize that they are *Subject Matter Experts (SMEs)* with regard to the effects of their disability. They were the experts on the use of relevant assistive devices, and the strategies that would best overcome the various barriers to learning. Since the student personas would be used by teaching staff, it was decided to incorporate the attributes initially used in the workshop proto-personas, and more explicitly link them to the structure of the teaching and learning encounters that students face. As a result, each persona speaks about their experience by 'commenting' on each of these areas as applicable. Of the different elements, three of these (*Lectures*, *Learning Materials* and *Exams/Accommodations*) are specifically linked to the educational sector and maps to key elements of the user journey of a student. The others can be considered core elements in a disability persona that can perhaps be used alongside domain-specific elements to build disability personas across a range of contexts. The resulting personas will be discussed in the following sections.

4 RESULTS

4.1 Teacher Personas

As described in Section 3.4, the teacher personas were produced to assist in the development of the student personas as well as the design of the hosting platform. Three teacher personas were developed to represent a broad range of teaching types. The *Averager* persona assesses themselves as having limited knowledge of digital accessibility and is unsure which of their teaching materials is accessible or not, and what the right methods are for improving accessibility. The *Minimiser* persona feels comfortable with the teaching methods that they have been using for some time, especially what they consider more traditional methods like using scanned materials, and writing on a whiteboard. They are often slow in learning and integrating new technology, and are often unwilling to modify their "tried and true" teaching methods materials. The *Optimiser* persona wants to or has tried to incorporate more accessibility into their teaching and learning materials, but they are often constrained by their lack of knowledge of best practices. They also often feel isolated and unsupported in their attempts to improve the accessibility of their teaching. These three teacher persona types were primarily used to assist with content decisions on what information would be most useful, and structural decisions on the best way to present the information in an engaging, interactive format.

4.2 Student Personas

Currently eight personas of students with disabilities are being developed for the platform. One example is Hannah, who introduces herself as a 25 year old undergraduate student majoring in Mathematics. Her flatmates and friendship circle include both hearing, and deaf and hard-of-hearing (DHH) students, and she enjoys playing handball, going out to eat and watching movies and TV

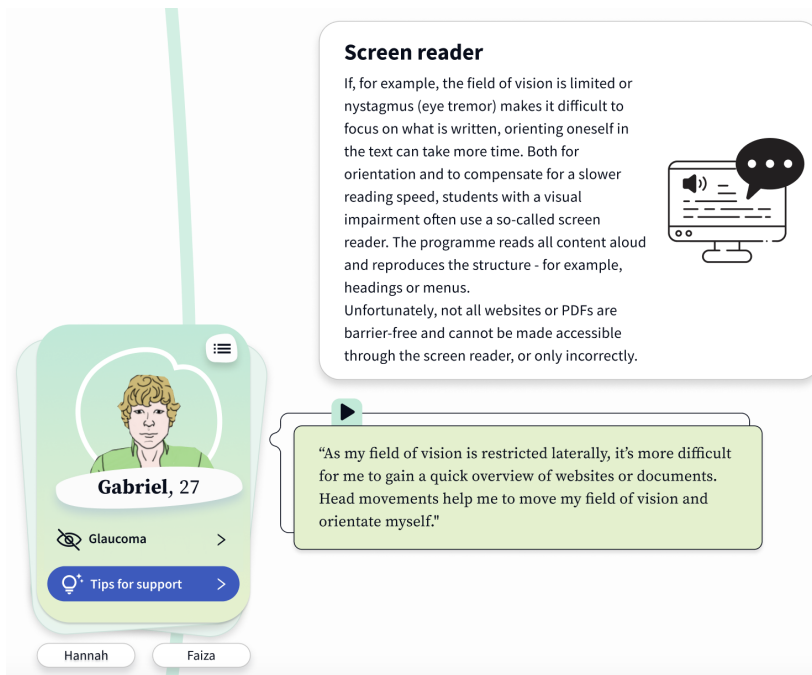


Figure 3: Prototype of the Gabriel Persona

series (Figure 2). Hannah describes her hearing disability, and her use of both a hearing aid and a cochlear implant. Hannah uses both spoken and signed language, but generally tends to prefer communicating in spoken language.

As seen as Table 1, Hannah comments on the content-based persona elements, presenting a richer experience in which she is the SME. She explains her disability, the types of assistive technology she uses, and what works particularly well for her in the various university contexts. She describes how lectures and tutorials with recorded captioned videos and audio descriptions can help her with her studies. She explains that live sessions should include a sign language interpreter if needed. Hannah points out that care should be taken to have optimal conditions for lip reading or usage of assistive listening devices especially in larger rooms, or in group discussions. She also provides insights into how information is not conveyed or conveyed inaccurately due to the presence of various barriers, and how this can contribute to feelings of exclusion. However, the focus is not on the negative aspects of Hannah's experience, but on her skills and strategies in navigating lectures, tutorials and exams, and letting the teaching staff know what she needs to help her succeed.

Another example persona is Gabriel (Figure 3), a student with a visual disability as a result of glaucoma. Gabriel is an undergraduate student in Social Work, a program with a significant reading component. Gabriel explains to the viewer that he can no longer see things that are at the edge of his visual field and can't see contrasts well either. He describes how he prefers to use a tablet for his work since it offers the freedom to enlarge texts or increase the contrast while reading, but points out how much of a slower reading speed he has. He relies a lot on screen readers for longer sessions,

but when linked websites and pdfs are not accessible, this is extremely frustrating. Like Hannah, Gabriel gives explanations and suggestions on how changes can be made to learning materials and teaching methods to make his learning experience more efficient and of a higher quality.

More personas are in various stages of the design process, including persona representations of students who are neurodivergent, as well as those with chronic illness, family care responsibilities, motor impairment, and psychological disorders.

5 FUTURE DEVELOPMENTS AND RESEARCH

This paper described the use of a co-design methodology to develop accessibility personas that could reduce barriers in learning for students with disabilities. Previous approaches to designing personas with accessibility have often fallen short and did not produce empathy or understanding in designers and developers. Moreover, the personas with disabilities were often portrayed as very one-dimensional, with the main focus on the disability and the resulting barriers the personas faces in a given environment.

Our aim was to develop approaches for creating personas that more authentically represent the experience of PwD in the higher education context. This approach is time-consuming and not suited to lean personas. The personas do not just give the user information about their disability but incorporate information about strategies, interests and points of view which help to portray PwD in a more multi-dimensional way. The co-design experience where PwD were involved in all stages of the design process – from making design decisions to feedback loops – helped ensure that the resulting personas portray authentic experiences. Further research will focus on evaluation of the personas and how effectively they actually

represent students with particular disabilities in order to validate the efficacy of the described approaches.

Additionally, development and evaluation will continue on incorporating interactive persona artefacts that help build a virtual encounter for target users. In the final stages, the persona artefacts will be evaluated *in situ* on the proposed web platform to examine how they are viewed and used by teaching staff, and how these virtual encounters change their attitudes and behaviours as they design and implement their teaching.

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